University of Sunderland

Network Planning & Design Assignment

CETM 51

250733064

1.0 Local Area Network Design

The Flies & Welders factory complex now has seven buildings and 121 devices, located at the end of various connections, spread across multiple floors. Given the business's rapid growth, the network architecture must support fault tolerance, scalability, quality of service (QoS), and security to operate efficiently and sustain growth (Cisco Systems, 2014).

Flies & Welders Inc. is now in the same area as the building design shows, but since they are an aerospace company, they might expand and have many sites around the world, with workers working from different places. Therefore, enterprise networks should be built to handle different kinds of traffic, such as important and combined data flows, meet various business requirements, and offer a central place for managing and controlling the network (Cisco Systems, 2014).

According to the Iso/IEC 11801 Ed. 2.2 (ISO/IEC, 2017) cabling standards, a hierarchical star topology is suggested because the Flies & Welders complex is a multi-building campus that has a logical central area (Cisco Systems, 2014). In a hierarchical model, the cables radiate from a central point to their final connections.

The Testing Centre, where the IT department, Systems Admin, and Systems Programmers are based, is the best place for the main equipment. A star-shaped design works well when there is a clear central point and several surrounding buildings that need to link back to that centre (Lammle, 2020). Usually, a star network setup can have issues with how far the connections can go. But if you use a ring or mesh setup instead, it will cost more, be harder to manage, and won't really help with how things work at this level.

A hierarchical network structure comprises three main parts: the core layer, the access layer, and the distribution layer. This model also aligns with the organisation's setup because the main component, located in the testing centre's IT room, will manage data flow across the company. Putting everything on a single switch means all decisions about how data moves are made in one place, making it easier to find and fix problems, manage access rules, and keep track of network traffic.

From a cabling perspective, the hierarchical star topology is almost always used because it's easy to set up and forms the foundation for international standards such as ANSI/TIA-568-C and ISO/IEC 11801 (TIA, 2018). It has a safe setup: if one workstation or its cable fails, the rest of the network keeps running, but if the main hub fails, it can bring the whole network down (Forouzan, 2013). Because the star layout has a central node, it's easier to find and fix problems compared to other designs. This is because each node can be checked and tested separately from the main connection point.

The distribution layer acts as a natural separation for each building (Cisco Systems, 2014). For example, if the switch in the Manufacturing Unit fails, only that building stops operating, while the rest of the site continues to function normally. The access layer is responsible for physically connecting the devices. Putting the floor-level switches apart from the distribution layer means that when you update or replace the access switches, it doesn't affect the core of the network. Also, as the company grows, you can add more access switches without completely redesigning

the system. In a growing environment, the hierarchical star topology works well because each level can expand independently (Odom, 2020).

To comply with telecommunications industry standards (TIA/EIA-568), connections between buildings must use fibre-optic cabling. Because the factory is large and there are connections between the indoor and outdoor areas, such as the deep pool, the distance between the buildings will likely exceed 100 meters, which is the maximum for copper cables. So, it's better to use fibre optic cabling to connect the core switch in the Testing Centre to the distribution switches in the other buildings. Single-mode cables are recommended for distances greater than 400 meters, or multi-mode cables for shorter distances.

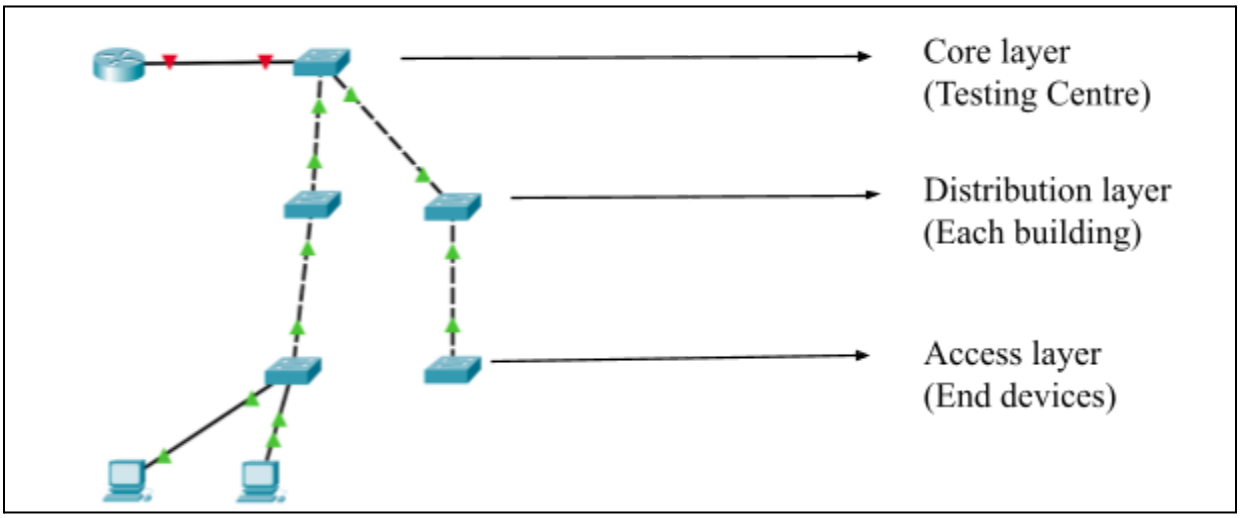


Fig 1. Local Area Network Design

1.1 IP System

For the Flies & Welders factory complex, using Internet Protocol Version 4 (IPV4) with the class B private addressing scheme is the suggested choice for the organisation because of the following reasons:

- IPV4 works with both older and newer computer equipment, as mentioned in Forouzan's 2013 work. At the Flies and Welders Manufacturing unit, the building houses PC-controlled industrial machines, such as precision lathes and robotic paint systems. Industrial Operational Technology (OT) systems and older manufacturing equipment usually lack IPv6 support, a newer Internet addressing system. This is noted in Tanenbaum and Wetherall's 2011 work. IPv4 helps the IT network used by businesses and the OT network used in manufacturing work together smoothly (Forouzan, 2013).

- The Class B network address range is suitable for medium- to large-scale organisations (RFC 1918, 1996), such as Flies & Welders. It can handle more than 65,000 devices, allowing the company to grow without issues.

- The private IP address range from 172.16.0.0/12 to 172.31.0.0/12 can support more than 1 million devices, as defined in RFC 1918 from 1996, which gives enough space for future growth.

- The Class B private IP address range, 172.16.x.x, is easy to spot because it's not used as often as the Class C private range, 192.168.1.0, which is commonly set as the default for home routers worldwide. Network engineers might find it simpler to recognise the 172.16.x.x range, which helps reduce confusion between personal devices at home and work servers for people working from home.

- The brief does not mention any services visible to the public, so there's no need for addresses accessible from anywhere on the internet. That's one of the main benefits of IPv6, as described in RFC 4291 (2006).

- IPv6 addresses, like fe80::1a2b:3c4d:5e6f:7g8h, are more difficult to read and verify in switch or router setups than IPv4 addresses, which use numbers separated by dots.

- Right now, the entire site needs fewer than 150 IP addresses across all endpoints. IPv6 was created to address the problem of running out of IP addresses worldwide, but this problem doesn't occur in small networks that use private IP addresses. An IPv4 Class B private subnet offers tens of thousands of available addresses, providing nearly unlimited expansion space for internal networks of this scale.

- Finally, the Network Address Translation (NAT) built into IPv4 systems naturally separates the internal factory network from the public internet, as described in RFC 3022 from 2001. Although IPv6 offers strong security features, IPv4 remains the preferred choice for local industrial setups that require simple management and quick fixes by on-site IT staff.

1.2 IP Scheme

Before designing the IP scheme, an audit of existing device requirements based on the site plan shows the following:

- Testing Centre: 55 PCs total
- Admin Building: 31 devices total
- Manufacturing Unit: 18 devices total
- Amenities Building: 10 devices total
- Warehouses (1 & 2): 4 devices total.
- Security Buildings: 3 devices total.

To create a good network and IP addressing system, the Flies and Welders Inc. network design has been divided into buildings, grouped by the departments they serve. This process, called subnetting (Lammle, 2020), is based on the number of devices in each building.

I will be using the Class B address block 172.16.0.0 with Variable Length Subnet Masks to ensure efficiency and minimise waste, as outlined in RFC 4632 (2006). This approach allows for scalability and supports growth in each building. This gives more than 1 million addresses available to support Flies & Welders Inc. with their needs now and in the future.

Building	**No of Hosts**	**Subnet bits**	**Host bits**	**Subnet ID**	**1st Host**	**Last Host**	**Broadcast**	**Subnet Mask**
Testing Centre	55	0	0000000	.0	.1	.126	.127	/25
Admin Building	31	10	000000	.128	.129	.190	.191	/26
Manufacturing Unit	18	110	00000	.192	.193	.222	.223	/27
Amenities	10	1110	0000	.224	.225	.238	.239	/28
Security	4	11111	000	.248	.249	.254	.255	/29
Warehouse 1	3	11110	000	.240	.241	.246	.247	/29
Warehouse 2	1	000000	00	.0	.1	.2	.3	/30

Table 1.0 Flies & Welders IPV4 Scheme

Even though only 55 hosts are needed for the Testing Centre right now, I upgraded to a /25 subnet mask to prepare for future growth and expansion. This area is the most important part of the building for the network setup. As a result, half of the available IP addresses in the range have been used, and Warehouse 2 has been relocated to the next set of addresses, 172.16.2.x.

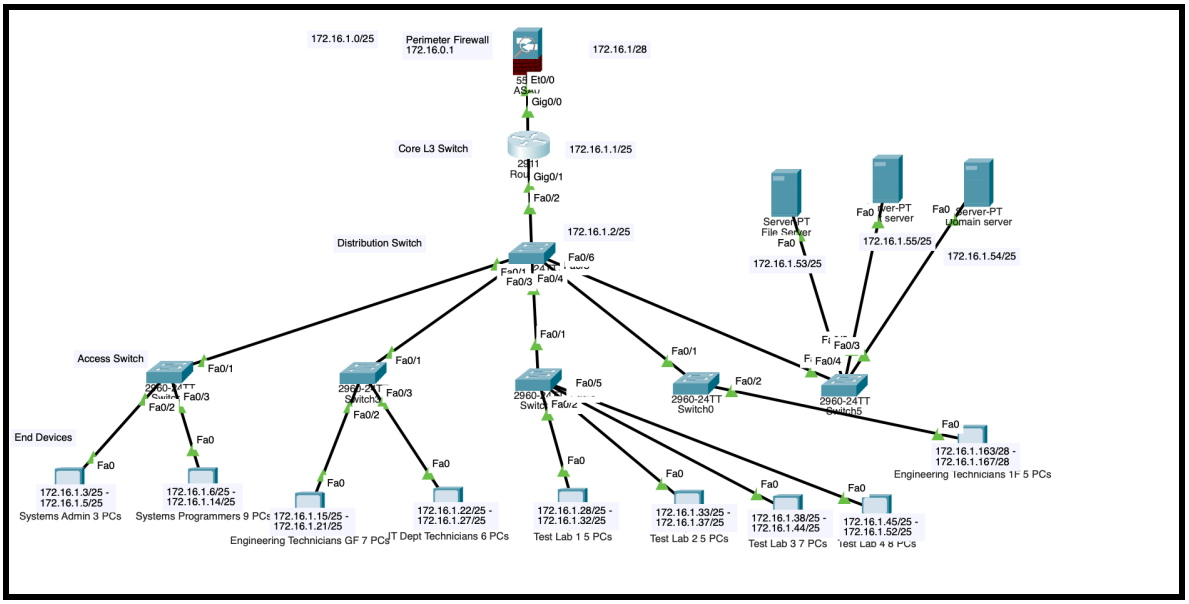


Figure 2. Testing Centre Network Diagram

The subnet for the testing centre is 172.16.1.0 with a subnet mask of 255.255.255.128. The engineering technicians on the first floor are connected to a separate subnet, 172.16.1/28, as shown in the network diagram. The router also has another connection, which lets it handle two separate networks. A firewall has been put on the ground floor. It is placed between the router and the outside networks. It acts as an extra layer of security, keeping the building safe from unwanted internet access.

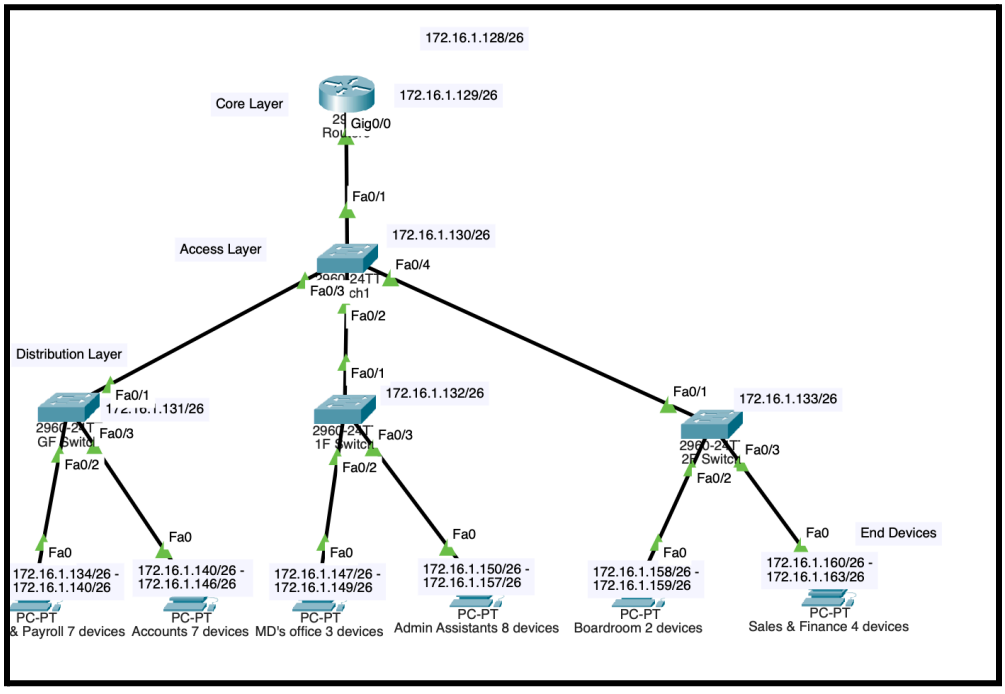


Figure 3. Admin Building Network Diagram

The subnet for the admin building is 172.16.1.128 with a subnet mask of /26. There is one subnet that includes all three floors of the building. On each floor, there is an access switch that connects back to the main central hub, which is part of the core layer. It's best to use vertical cabling that goes up the building.

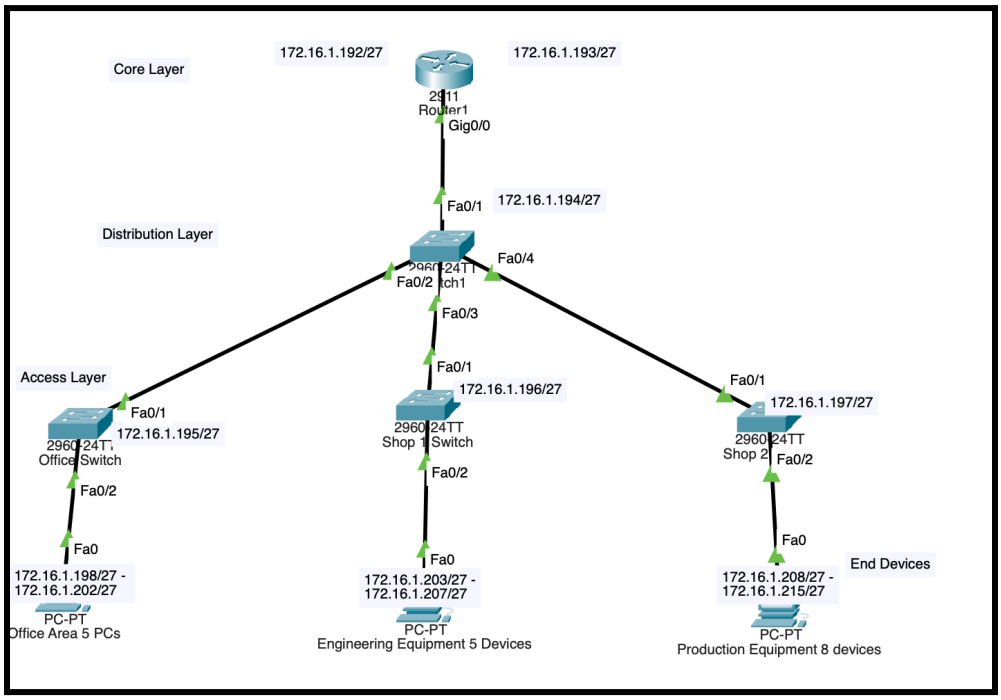


Figure 4. Manufacturing Unit Network Diagram

The subnet address for the Manufacturing Unit is 172.16.1.192/27. There are three access switches placed in the office area, shop 1, and shop 2.

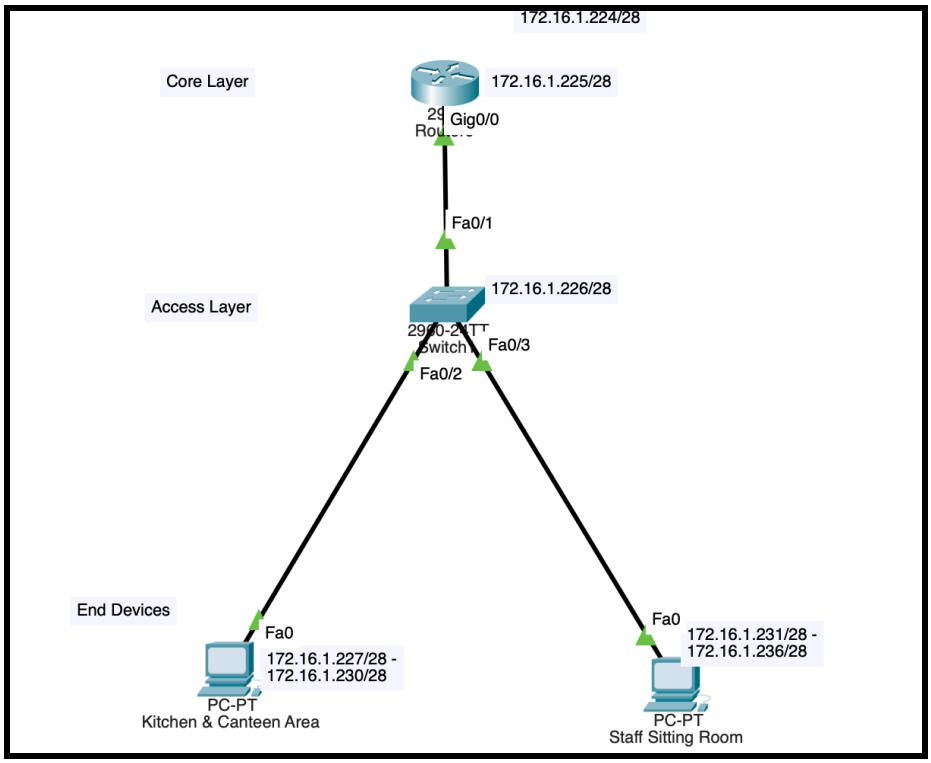


Figure 5.Amenities Building Network Diagram

The subnet address for the Amenities Building is 172.16.1.224 with a 28-bit subnet mask. Because there are not many hosts or end devices in this building, I set up a two-tier hierarchical network structure. This method helps reduce costs and make the network more reliable by reducing points of failure. The distribution switch is directly connected to the end devices. There are six computers in the staff sitting room that are used for browsing the internet. These computers will not be allowed to connect to other parts of the internal network. This is done through an Access Control List (ACL) set up on the core router. This helps ensure guests can't see other private information on the network, such as the POS machines in the staff canteen.

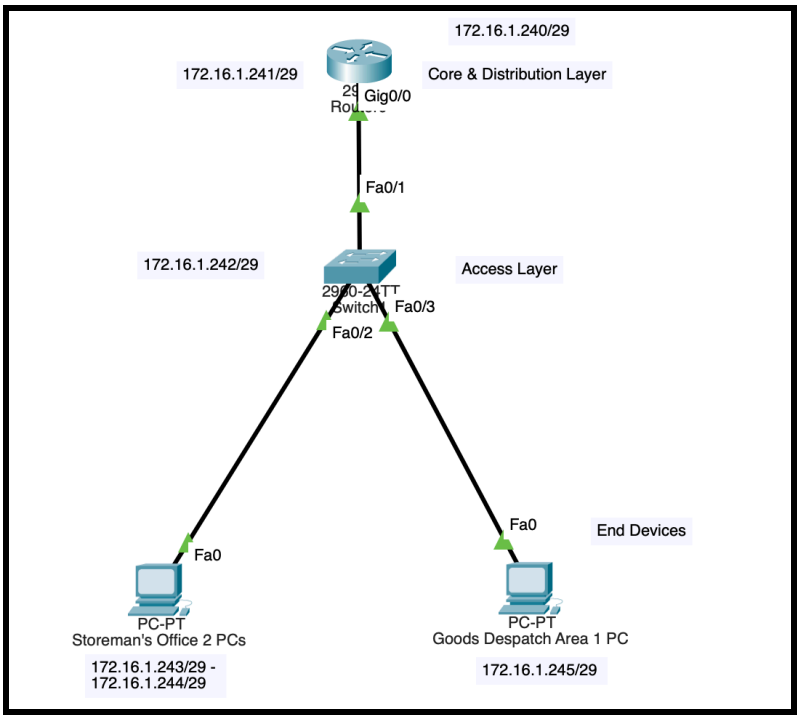


Figure 6. Warehouse 1 Network Diagram

The subnet address for Warehouse 1 is 172.16.1.240/29. Like the Amenities building, the end devices connect to the distribution switch, which is connected directly to the core router. This setup doesn't require a distribution switch since the building has only a small number of devices.

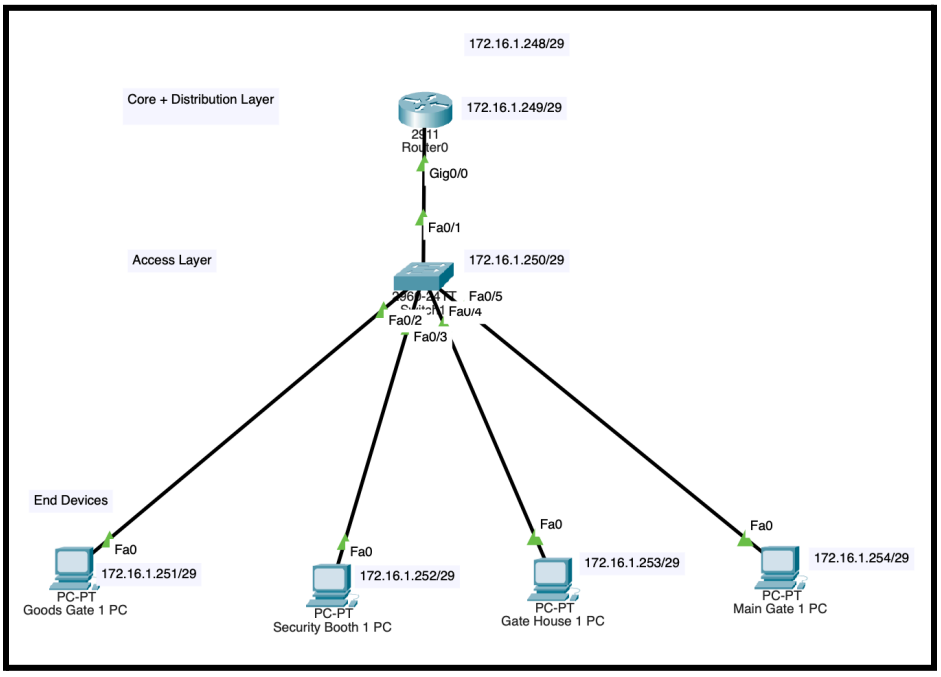


Figure 7. Security Buildings Network Diagram

The subnet address for the Security Building is 172.16.1.248 with a subnet mask of /29. As with the Amenities building, the end devices connect directly to the distribution switch that connects to the core router. This setup doesn't require an access switch because the building has very few devices.

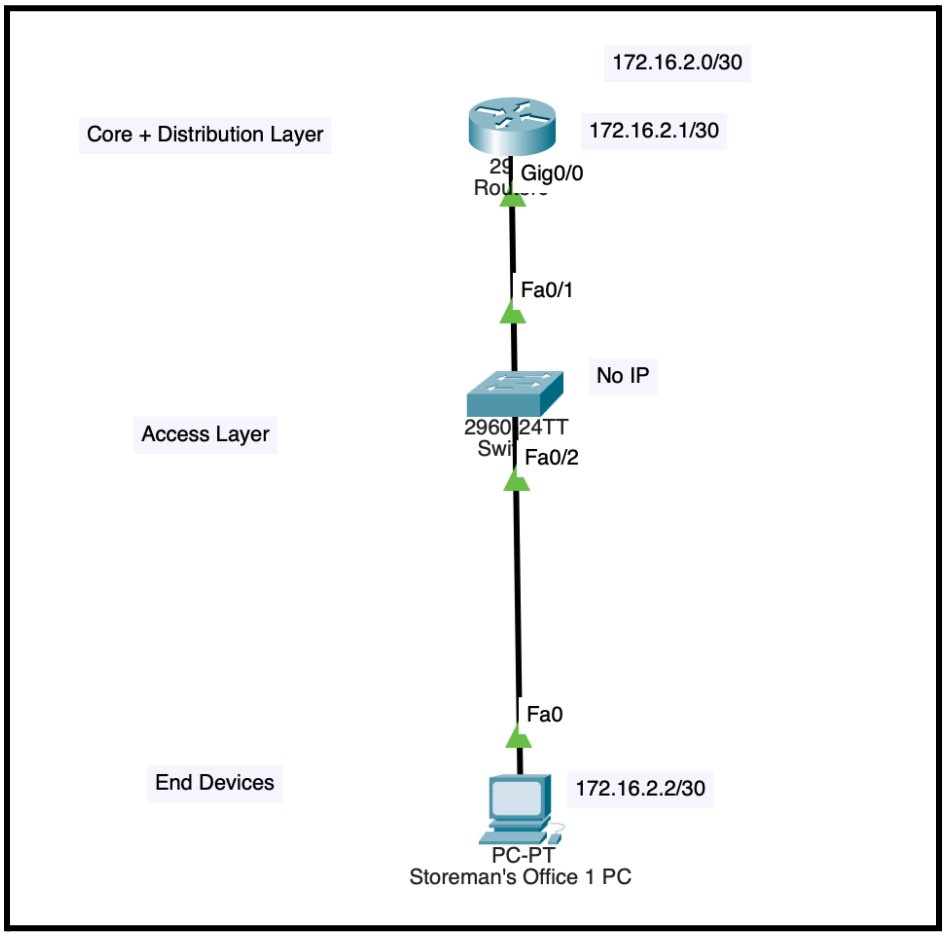


Figure 8. Warehouse 2 Network Diagram

The subnet address for Warehouse 2 is 172.16.2.0 with a subnet mask of /30. Because all the addresses in the 172.16.1.x range has been used up, I have relocated Warehouse 2 to the 172.16.2.x address range. In addition, a /30 subnet mask allows for two usable IP addresses: one for the router and the other for the end device. This can still work because the building has very low hosting needs. In this setup, the switch will work just as a simple device that forwards data without being able to receive messages or be reached by other devices on the network. One advantage of this is that it reduces the available surface for attack on the network. In the future, if another computer or device is added to this building, it can be moved to the /29 subnet mask, which might affect the entire IP address plan.

2.0 Network Design

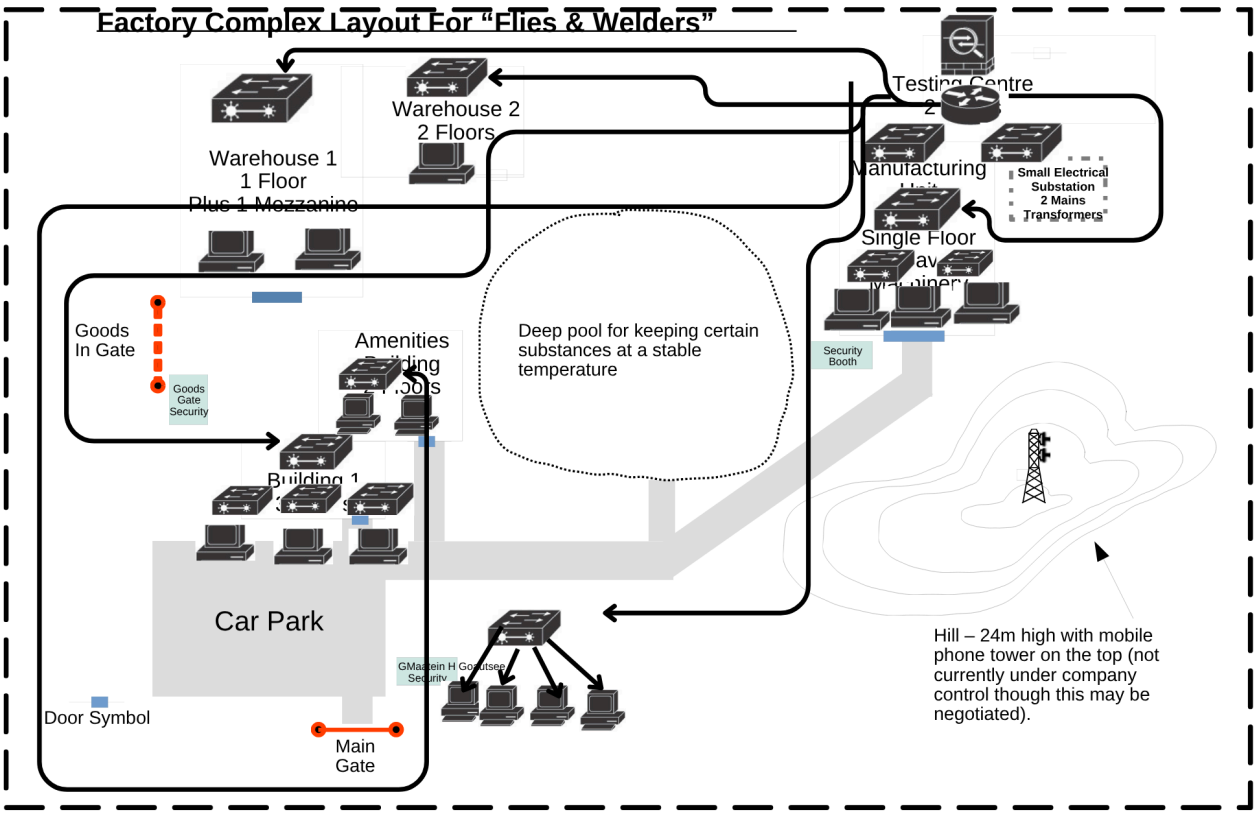


Figure 9. Flies & Welders Network Design Plan

2.2 Issues with Site Features

Some of the limitations to take into account while designing the network plan for Flies and Welders Inc. include:

Deep pool: The deep pool is located right in the middle of the factory area. There are inherent risks in running cables in a pool due to potential contact with harmful substances; the cabling system will need to be installed around the pool. The extra distance will also change how much cable is required in the network plan.

Hill: In the eastern part of the area, there is a 24-meter-high hill, and on top of it stands a mobile phone tower. Even though the hill is high and makes it harder to run cables and move around, it would be helpful if the company could use it later. That's because it could be used to transmit wireless wide-area network signals to the internet service provider. For the current design, fibre optic cable will be buried at the base of the hill.

Small electrical substation: There is a small electrical substation situated on the right side of the complex, right next to the manufacturing unit. Even though it's a small substation, it has two main transformers, which might generate significant electromagnetic interference (EMI) through coupling. Because of this, the cables must remain more than 1 meter from the substation, as specified in EN 50174-2. Using fibre optic cables makes them more resistant to electromagnetic interference.

Car park: There is a car parking area situated at the southern part of the building complex, near the administration building. Cars or other vehicles might cause wear or damage to the cables. When laying cables, it's important to remember that they should be placed at the right depth under the car park surface. According to TIA standards, cables must be buried at a minimum depth of 750mm, and high-density polyethene (HDPE) concrete protection slabs should be placed on top (TIA-569-D, 2015). Drawstrings will be added to allow for more cores to be added later.

Gates: Located on the south and west sides of the complex. The main gate is on the south side, and the goods gate is on the west side. Once again, vehicle traffic causes the cables to wear out and become damaged. Because of this, cables must be placed beneath roads in accordance with the 2015 TIA-569-D standard. Pre-installed HDPE ducted crossings will be buried at a minimum depth of 600 meters under the gates, with a focus on accommodating heavy vehicles.

Inter-building: Since it's a big factory complex, the distance between the buildings needs to be considered. According to TIA standards, all cables between buildings that run over paved or busy areas must be placed in an HDPE conduit buried underground. The minimum depth for this conduit is 450 millimetres under walking paths, 600 millimetres under roads, and 750 millimeters under parking areas. This follows the guidelines of BS 7671:18, published by the IET in 2018.

2.3 Recommended Media

Three main types of cables are recommended for the Flies & Welders factory complex:

A. OM4 Multimode Fibre Optic Cable

For the campus area network (CAN), it's best to use a multimode fibre-optic cable in the network setup. This cable will act as the main connection throughout the factory complex. It starts from the Testing Centre IT Department Room and links the distribution switches located in each building across the complex.

As mentioned earlier, multimode fibre-optic cables can handle speeds up to 10 Gbps over distances longer than 100 meters (IEEE, 2002), making them a good choice for the building-to-building setup here. In addition, fibre is immune to EMI. Because there's a substation on site, fibre-optic cable will be the preferred option. The cable run will have at least two extra wires in the duct to allow for adding more cables later.

Multimode fibre optic cabling, specifically OM4, was selected instead of single-mode OS2 because the cabling requirements are limited to a single location and do not require support for distances longer than 400 meters. OS2 fibres can transmit data over longer distances, but they require more expensive transceivers. Because of this, OM4 is a better and cheaper option right now.

B. Cat6a (Unshielded Twisted Pair) UTP

For vertical and horizontal cabling inside buildings, it is recommended to use Cat6a (Unshielded Twisted Pair) cables in accordance with the TIA-568.2-D standard (TIA, 2018). Cat6a cables can handle speeds of up to 10 Gbps over 100 meters, according to TIA in 2018, which is better than regular Cat6 cables. These are good to use in places where there's no EMI issue and can be used in regular office and administrative work areas.

For the manufacturing building where the machines are located, it is best to use copper cables that are either foiled or unshielded twisted pairs (F/UTP) to reduce interference from the CNC machines. F/UTP cables are simpler to connect with RJ45 plugs.

C. Armoured Cat6A

For the different security buildings, it's recommended to use armoured Cat6A cables because the steel covering provides an extra layer of protection on the outside. The cables in these areas are often hit by vehicles. Also, the buildings are located in areas with less wiring, yet they still offer speeds of up to 10 Gbps.

2.4 Expanded IP Scheme

An expansion of the IP scheme from part 1 that covers the entire factory complex is shown below. Because all the IP addresses in the 172.16.1.x range has been used up, the infrastructure's IP addresses will be changed to the 172.16.2.x range to make them different from the others. The IP system has mostly been divided into building subnet IDs and infrastructure subnet IDs to lower the chance of address conflicts or overlaps. It's important to note that the warehouse 2 subnet ID is already included in this scheme.

The infrastructure link covers the wide-area network, backbone, and network-management connections for each building. The expanded IP scheme also includes information about unused IDs from the 172.16.1.x address range that can be used within the buildings to support future growth and expansion. The Testing Centre has the most spare IP addresses because it contains much of the factory's technology and network setup.

The 172.16.2.x range of addresses has been carefully assigned to minimise unused address space in each subnet. There are still more than 150 unused IP addresses from 172.16.2.76 to 172.16.2.255 available, just in case a new building or part of the factory complex needs to be set up.

Building	No of Hosts	Subnet bits	Host bits	Subnet ID	1st Host	Last Host	Broadcast	Subnet Mask	Spare IPs
Testing Centre	55	0	0000000	.0	.1	.126	.127	/25	71
Admin Building	31	10	000000	.128	.129	.190	.191	/26	31
Manufacturing Unit	18	110	00000	.192	.193	.222	.223	/27	12
Amenities	10	1110	0000	.224	.225	.238	.239	/28	4
Warehouse 1	3	11110	000	.240	.241	.246	.247	/29	3
Security	3	11111	000	.248	.249	.254	.255	/29	3
Warehouse 2	2	000000	00	.0	.1	.2	.3	/30	0
ISP Link 1	2	000001	00	.4	.5	.6	.7	/30	0
ISP Link 2 (redundant)	2	000010	00	.8	.9	.10	.11	/30	0
Core Mgt	6	00100	000	.12	.13	.18	.19	/29	0
Campus Backbone	14	0100	0000	.20	.21	.34	.35	/28	0
Network Mgt	30	100	00000	.36	.37	.66	.67	/27	0
Redundant link	6	11000	000	.68	.69	.74	.75	/29	0

Table 2. Expanded Network IP Scheme

2.5 Redundancy Features

Based on the TIA-568.3-D and TIA-569-D structured cabling standards from 2015 and 2016, the network design includes separate fibre paths for the backbone connection, which is important for having redundancy. This means that if there's a problem in one part of the network path between the testing centre, the manufacturing unit, or the admin building, such as a router's power supply failing or a cable being damaged, the rest of the network can still stay connected.

The deep pool in the middle of the complex is a source of redundancy concern because if something happens to one of the backbone cables, the whole network could go down. Hence, the cables have to go around the water area so that if one cable is damaged, the rest can still work.

By using two ISP routers, two separate internet connections are established, providing a backup system for the Wireless Area Network (WAN). Also, the mobile mast on the hill can serve as an additional source of redundancy for the WAN by housing microwave transmitters. WAN connections are usually the most complex part of a network and are more likely to fail due to individual components.

If the budget allows, it is also recommended to have a redundant core L3 switch in the main hub; this helps remove the core router as the only weak point in the network setup. This has also been considered in the expanded IP scheme.

UPS (Uninterruptible Power Supply) devices will be installed in the IT room to ensure the core router and firewall continue operating during a power outage. The factory complex includes a substation that has two main transformers. The central hub will be linked to these transformers so that if one fails, the other can keep the network running.

In each building, an L2 switch would also be connected to a UPS-powered circuit, which is especially recommended for security buildings to ensure security in the factory complex is always maintained.

By enabling the Rapid Spanning Tree Protocol (RSTP) on all managed switches, a broadcast storm can be prevented if an accidental loop forms, as noted in IEEE's 2004 work. This is also a required component per structured cabling standards.

2.6 ISP Connections

The site plan indicates that six computers in the staff room will be used for browsing, which is why internet service providers (ISPs) are included in the network design. In the updated network design, the Testing Centre IT room now includes two ISP connections. Two ISPs are preferred over one, so that if one goes down, the other can still work.

Both ISP connections will end at the perimeter firewall located between the router and the WAN. The firewall acts as the boundary between the internal network and the internet; ISP connections must terminate at this point. The firewall is connected to the core router, and traffic from the router will be routed to other parts of the building.

By placing the ISPs and other key network devices in the Testing Centre, the IT team can easily monitor and manage them. In the future, if the facility has greater internet browsing needs, both ISPs can be used together to share internet traffic, doubling the organisation's available internet bandwidth.

3.0 Cost Estimate

Prices have been sourced from various UK retailers and include 20% VAT. In addition, cost estimates are exclusive of Consultants' fees and labour installation costs.

Category	Item Description	Justification	Location	Qty	Unit cost (£)	Total Cost (£)
Network Devices						
Firewall	Fortinet FortiGate 80F NextGen	The FortiGate 80F can handle a maximum firewall speed of 10 GBps. It has dual-WAN ports that connect to two ISPs and supports NAT, VPN, and ACL settings. These features are important for controlling which parts of the network the meeting room staff can access via their web browsers. Costs significantly less than Cisco's ASA 5525-X, which is around £4500, for the same features. FortiOS is also ICSA-certified.	Testing Centre - Systems Admin office	1	820	820
Core router	Ubiquiti UniFi Dream Machine SE	UDM-SE includes L3 managed switching, inter-VLAN routing, DHCP, DNS, VPN endpoint, and a network controller that works with all 15 UniFi switches. It costs much less than Cisco's Catalyst 9300-48P-E, which is about £3368, to do the same things.	Testing centre - Systems Admin office	1	650	650

		All the connected distribution and access switches can be controlled from a single web-based dashboard. Supports RSTP, STP, and LACP protocols, along with 10G SFP+ backbone uplinks, which are needed for redundancy.				
Distribution switch	Ubiquiti UniFi USW-Pro-24	The USW-Pro-24 offers 24 Gigabit Ethernet ports and two 10-gigabit SFP+ uplinks, allowing it to connect to an OM4 fibre backbone in the main distribution frame. It has a centrally controlled L3 switch with a touchscreen LCD display for checking the local status, which works really well for Flies & Welders' spread-out locations. This switch supports VLANs, LACP 802.3ad, STP and RSTP 802.1D/w, and 802.1X port security, all of which are necessary for a hierarchical star network topology. It costs much less than Cisco's 2960X, which is around £600 per unit, to do the same things.	All buildings	7	355	2,485
Access switch	Ubiquiti UniFi USW-48	A 48-port managed switch is needed to manage 55 devices in the testing centre without daisy-chaining, which could cause delays and create single points of failure in the network.	Testing centre - ground floor (55 devices)	1	425	425

		Suitable for use in an IT office or lab setting, with no fan.				
Access switch	Ubiquiti UniFi USW-24	The 24-port managed switch has enough space to grow as the testing centre expands.	Testing Centre, 1st floor cabinet, serving 5× Engineering Technician PCs on subnet 172.16.1.160/28.	1	195	195
Access switch	TP-Link TL-SG3428XMP JetStream	The TP-Link JetStream managed PoE switch offers both data connection and electrical power. It was selected for the Manufacturing Unit instead of Ubiquiti because the strong EMI from the CNC machines in Shops 1 and 2 could interfere with features that rely on wireless connections. Also, the metal casing makes it more suitable for industrial use.	Manufacturing unit	3	280	840
Access switch	Ubiquiti UniFi USW-24	The UniFi USW-24 is a 24-port managed switch. One switch is required on each floor to meet TIA-568 horizontal cabling standards. The switch supports 802.1X port-based authentication, meaning devices must prove they are authorised to connect before they can use the network. This works best for the Admin building, where the directors' and boardroom computers are located. In the Amenities building, the switch will be configured with VLANs to place the staff's internet browsing stations in their own separate VLAN.	Admin building - 1 per floor Amenities building	4	195	780

		This is important to keep the company's private data safe from guests.				
Access switch	NETGEAR GS308PP 8-Port PoE+ Unmanaged Switch	A 24-port switch is not a good choice for the security building because only 4 devices need to connect. The 8-port NETGEAR GS308PP offers PoE+, providing 30 watts per port (83 watts total) to power IP cameras and door access control readers at every gate post, with no need for additional power cables.	Security building	1	115	115
Servers						
File server	Dell PowerEdge T150 Tower Server Xeon E-2314, 16GB ECC, 2×2TB HDD RAID-1	The file server needs more storage space because it offers network shared drives used by all departments. RAID-1 mirroring protects against single drive failure. Dell's 3-year ProSupport warranty can be added as an extra option.	Testing Centre, 1st floor	1	895	895
Domain server	Dell PowerEdge T150 Tower Server Xeon E-2314, 16GB ECC, 1TB SSD	The Dell PowerEdge T150 Tower Server includes DNS and DHCP services, and it runs Windows Server 2022 Essentials to manage Active Directory, handle DNS lookups across all subnets, and assign IP addresses dynamically within each subnet's IP range. It's important to keep the domain server and the file server separate so that domain services continue to	Testing Centre, 1st floor	1	895	895

		function even if the file server needs to be restarted for maintenance.				
Print server	Dell PowerEdge T150 Tower Server Xeon E-2314, 8GB ECC, 512GB SSD	The Dell PowerEdge T150 Tower Server manages all the network printers in the Admin Building from a single central location, and it's set up to easily add more printers later if needed.	Testing Centre, 1st floor	1	799	799
Cabling & Trunking						
OM4 Multimode Fibre Cable	12-core LSZH 50/125µm OM4 loose tube, armoured outdoor grade	Supports 10GbE to 400m per IEEE 802.3ae. Minimum order is 20m.	Inter-building backbone runs buried in HDPE conduit.	~1200m	3.80/m	4,560
Straight through copper cable	Cat6A UTP Cable 305m Box Excel Cat6A LS0H CPR-B2ca rated	Cat6A unshielded UTP cable has a B2ca CPR fire rating and comes in 305-meter boxes. It supports 10GbE up to 100 meters.	All buildings have standard horizontal cabling connecting the access switches to wall outlets in office areas, labs, the canteen, and warehouses.	~3050m (10 boxes)	193	1,930
Straight through copper cable	Cat6A FTP Shielded Cable 305m Drum Labgear Cat6A FTP LSZH	Cat6A cable with foil shielding (FTP), includes a drain wire, made with LSZH material, and comes on a 305m drum. Designed for high-interference environments per IEC 61000-4. All CNC lathes and robot paint units require shielded cables because the high-current motors and drive electronics generate significant electromagnetic interference.	Manufacturing Unit Machine Shop 1 and Shop 2 1 additional drum for the security perimeter armoured runs between the Gate House, Goods Gate, and Main Gate	3 drums (915m)	245	735

Steel Conduit 20mm, heavy gauge galvanised	20mm galvanised steel conduit, 3m lengths with couplers and bends.	All Cat6A FTP cables are run through steel conduit to provide the best protection against electromagnetic interference in line with the IEC 61000-4 standards. The conduit is kept at least 300 millimetres away from high-voltage supply cables, as specified in EN 50174-2 Table 4.	Manufacturing Unit - Machine Shops 1 and 2.	60 lengths (180m)	4.80/length	288
HDPE Twin-Wall Conduit	50mm HDPE twin-wall corrugated duct, draw cord pre-installed.	Suitable for direct burial per EN 50086-2-4. Minimum 2 spare cores per run. Draw cords should be kept in the conduit so that additional cables can be added later without excavating the area.	All inter-building outdoor fibre routes. The cables are buried at a depth of 450mm as standard, or 600mm in areas such as car parks and access roads, in accordance with the guidelines in BS 7671:2018 and EN 50174-2:2018.	800m	2.20/m	1,760
Patch Leads	OM4 LC-LC Duplex Fibre Patch Leads - 2m	Centralised organisation for multiple cables.	Core router and all distribution switch locations connect fibre patch panels to SFP+ ports on the switches. There are two per location, one primary and one spare, for 9 locations, plus extra ones.	30	7.5	225
Patch Leads	Cat6A Cat6A UTP RJ45 Patch Leads - 0.5m and 1m assorted	Different colours are used to identify VLANs. The cable lengths are 0.5 meters from the switch to the panel and 1 meter from the panel to the device. Colour codes are used to show different types of access. Blue means	All buildings, the connections between patch panels and switches in every core and distribution cabinet, and the cables running from the wall outlet to the devices.	200	2.50	500

		standard access (VLAN). Red means security access. Yellow means restricted access for browsing amenities stations.				
Steel Cable Tray	Perforated steel cable tray, 100×50mm, 3m lengths with joiners, bends, and wall brackets. 100mm wide for horizontal cabling.	Maintains separation between data and power cables in accordance with EN 50174-2. Allows easy access to structured cabling for maintenance and troubleshooting.	All buildings have cable management systems that run from the distribution switch cabinets to the floor or wall distribution outlets.	80 lengths (240m)	12.50/length	1,000
Fire-Rated Cable Tray	Fire-rated steel cable tray, 150mm wide, 3m lengths for vertical riser cable routes between floors.	A fire-rated tray is required according to BS 7671:2018 Regulation 527 for cable passages in buildings with more than one floor. Helps prevent fires from spreading up or down between floors via cable paths.	Admin Building (3-storey) and Amenities Building (2-storey)	20 lengths (60m)	22.00/length	440
Network Racks & Cabinets						
Equipment cabinet	22U Floor-Standing Server Rack	19-inch, locking perforated front and rear doors, 800mm depth, casters, PDU-ready, fan-cooled. EIA-310-D compliant.	Testing Centre IT room. The setup includes a UDM-SE, a FortiGate, a 48-port TC GF switch, three servers, a UPS, patch panels, and a cable management system.	1	380	380
Equipment cabinet	12U Wall-Mount Cabinet	Standard 12U wall mount with a depth of 600mm, 19-inch size, includes a locking glass door, two 120mm fans, removable side panels, and wall-mount brackets. Load capacity 60kg.	Warehouse 1: ground floor, Warehouse 2: ground floor, Admin Building: ground floor Amenities Building: ground floor	6	185	1,110

			Gate House Security area, and the Testing Centre: 1st floor, which houses the access switch.			
Equipment cabinet	12U Wall-Mount Cabinet - IP54 Rated Industrial	12U, 19-inch, IP54 waterproof/dustproof enclosure. Designed for use in industrial settings where there may be metal dust or liquid splashes, meeting the standards set by IEC 60529. The IP54 rating helps safeguard the switch equipment against metal shavings, coolant spray, and other dust and debris that come from CNC machines.	Manufacturing unit with a separate cabinet for the access switches of Machine Shop 1 and Shop 2.	1	295	295
Equipment cabinet	16U Wall-Mount Cabinet	16U, 19-inch, 600mm depth, lockable. The space is slightly larger to accommodate the distribution switch, office access switch, and patch panels in the Manufacturing Unit office area.	The manufacturing unit's office area has a cabinet that contains the USW-Pro-24 distribution switch, an office-area TP-Link access switch, patch panels, and a UPS.	1	220	220
Patch Panels, Sockets & Transceivers						
Patch panel	24-Port Cat6A UTP Patch Panel	24-port Cat6A cable with 110-IDC rear termination, meets TIA-568-C.2 standards, includes a rear cable management bar, and has dual T568A/B labelling.	All buildings: (Testing Centre x2, Admin building x3 and Manufacturing Unit x2).	14	52	728
Patch panel	12-Port Fibre Patch Enclosure - LC OM4	A 1U rack or wall mount fibre enclosure that can take pre-terminated OM4 pigtails or allow for field terminations.	Core router and all seven distribution switches, plus one spare.	9	45	405

		Each enclosure connects the incoming OM4 backbone fibres and allows for LC-to-LC connections to the SFP+ ports on the neighbouring switch.				
Wall plate	Cat6A Double Keystone Wall Plate + Backbox, white.	A UK double-gang faceplate with two Cat6A keystone ports and two RJ45 ports. High performance, allowing speeds of up to 10 gigabits for 100 meters and offering 500MHz bandwidth. Includes surface or flush backbox.	Site-wide. All office areas, labs, and workstations, including the spare ones.	~ 120	5.80	696
Cat6A Shielded Keystone Module	TUK SGKSJE die-cast Cat6A STP keystone Shielded Cat6A keystone, die-cast zinc housing, tool-less butterfly termination. TIA-568-C.2 compliant..	Shielded keystones are required to keep the entire path from the patch panel to the CNC device fully shielded; if the shield at the keystone is damaged, the FTP cable will no longer protect against electromagnetic interference.	Manufacturing Unit Machine Shops 1 and 2.	40	8.50	340
Cat6A UTP Keystone Module	Cat6A UTP keystone, tool-less IDC, 110-type, compatible with standard UK faceplates and patch panels.	This is a standard module that works with unshielded Cat6A cables. These cables are simpler and more flexible to install than shielded versions like FTP or STP.	In all areas of every building that are not EMI-impacted, one keystone should be placed at each outlet port on all faceplates and patch panels.	200	3.50	700
Transceiver	SFP+ 10GbE Multimode Transceiver (OM4)	SFP+ 10 Gigabit transceiver, uses LC connectors, operates at 850nm wavelength, and can transmit data up to 400 meters on OM4 fibre. Works with Ubiquiti, TP-Link, and most enterprise switches.	All switches with SFP+ uplink ports.	20	18	360

Power Devices						
UPS	APC Smart-UPS SMT 1500VA 2U Rackmount	APC SmartConnect offers cloud-based monitoring, comes with a 2-year warranty, and has a runtime of about 15 minutes when operating at full load. Helps protect UDM-SE, FortiGate, patch equipment, and servers. It runs for 15 minutes, which is enough time to safely shut down servers or start a generator. SmartConnect sends email alerts on power events.	Testing Centre IT room	1	872	872
UPS	APC Back-UPS 650VA desktop UPS	The setup takes about 8 minutes to run at 50% load, which is important for distributing switches, setting up access control devices at the Security Gatehouse, and ensuring network continuity for the CNC machines in the manufacturing unit.	Admin Building equipment cabinet (x1) Security Gate House cabinet (x1) Manufacturing Unit cabinet (x1)	3	68	204

Category	**Cost (£)**
Network devices	6,310
Servers	2,589
Cabling & Trunking	11,438
Racks & cabinets	2,005
Patch panels & sockets	3,229
Power Devices	1,076
Grand total	**26,647**

Table 3. Cost Estimate for Network Plan

4.0 PC Design

According to Autodesk's official system requirements for AutoCAD 2026 (Autodesk, 2026), processor speed, memory, and video card graphics memory are more important than the specific brand of these components. In addition, the £2,000 per-unit limit encourages a custom-made, unbranded workstation as the most cost-efficient option for the Researchers' PC Design.

AutoCAD 2026 is recommended over earlier versions. Autodesk offers a subscription model, and with a full AutoCAD 2026 subscription, users can access the three latest versions. This allows them to try out different versions on their computers and ensure their designs work well with various software versions.

Key reasons for choosing AutoCAD 2026 full over AutoCAD LT include:

- AutoCAD LT is only for 2D work and lacks 3D features, which can make it harder for researchers to perform their tasks.
- Full AutoCAD includes 3D modelling features, tools for creating solid and surface models, and mesh tools, all of which are important for an aerospace company.
- It has seven different toolsets tailored for specific industries, and the Mechanical toolset is particularly useful for designing and testing components.
- The 2026 version includes AI-powered Markup Import & Assist, Autodesk Assistant, and Smart Blocks, but these features are not available in AutoCAD LT.

All prices are taken from UK-based stores and do not include VAT.

4.1 PC Components

Internal Essentials					
Component	**Manufacturer/Model**	**Justification**	**Qty**	**Unit Cost Minus Tax (£)**	**Number Required (8) x Unit Cost (£)**
Motherboard	ASUS Prime B760M-A WiFi DDR5 (Intel LGA1700, mATX, B760, PCIe 4.0, 2× M.2, 2.5GbE, WiFi 6, HDMI + DP	The LGA1700 socket is compatible with Intel's 12th-, 13th-, and 14th-generation processors, making it possible to upgrade to newer CPUs in the future without replacing the motherboard. DDR5 memory offers much faster data transfer speeds than DDR4, which is especially useful when working with large DWG files and complex 3D models.	1	140	1,120
CPU	Intel Core i7-14700 (14th Gen, 20 cores 8P+12E, 5.4 GHz boost, 65 W TDP, LGA1700, boxed inc. Laminar RM1 cooler)	High-performance cores provide the high single-core InterProcess Communication (IPC) that AutoCAD requires. Autodesk suggests a minimum of 3 GHz for general use, but for 3D work, they recommend a turbo boost of 4 GHz or higher. The i7-14700 satisfies both.	1	295	2,360
RAM	Corsair Vengeance DDR5 32 GB (2× 16 GB) 5600 MHz CL36, dual-channel	Meets Autodesk's 32 GB recommendation, which is enough for professional 3D work. Using dual memory channels increases memory bandwidth, which is important when working with large assembly drawings.	1	88	704
Storage	Samsung 980 Pro 1 TB NVMe PCIe 4.0 M.2	AutoCAD 2026 full requires about 16-20 GB of space; 1 TB can hold project archives, DWG files,	1	82	656

		and the operating system.AutoCAD's performance during loading is closely linked to the storage device's speed. A mechanical hard drive is not included because a separate hard drive is not needed. The Testing Centre already has network-shared file storage configured via the domain controller and file server.			
System Case	Fractal Design Focus 2 ATX mid-tower (tool-free, no RGB, good airflow, unbranded appearance)	The airflow system design ensures the i7-14700 and RTX 4060 stay within safe temperatures without generating excessive fan noise. In a professional research setting, using RGB lighting is not suitable.	1	62	496
Power Supply Unit	be quiet! System Power 10 650 W 80+ Bronze ATX	650 W offers enough extra power to last longer and helps prevent the power supply unit from working too hard. An efficiency rating of 80 or higher means the product uses energy well and produces less heat.	1	64	512
GPU/Display Card	Nvidia MSI GeForce RTX 4060 Ventus 2X OC 8 GB GDDR6 (DirectX 12, 2× DP + 2× HDMI, 272 GB/s bandwidth)	8 GB of GPU memory meets the recommended requirement for AutoCAD 2026. Having 8 GB of VRAM is enough to render 3D models at Quad High Definition (QHD) resolution on two screens simultaneously. The RTX 4060 is the best option within the budget, compared to the NVIDIA RTX A2000, which is a certified workstation card and costs £370 or more. Autodesk supports certain hardware that's certified for warranty reasons, but the RTX 4060 works properly in AutoCAD.	1	279	2,232

		The RTX 4060 is also built on the NVIDIA Ada Lovelace architecture and has a 115W power draw, making it very energy-efficient. The 650 W power supply unit provides extra capacity to run it properly.			
Network Interface Card (NIC)	Not required	The ASUS Prime B760M-A WiFi DDR5 has a built-in Realtek 2.5 Gb Ethernet port that can connect directly to the Testing Centre network at 2.5 Gbps, so there is no need for a separate network card.	0	0	0
Total				£1,010	**£8,080**

Table 4. Internal Essentials for PC Design

External Essentials					
Component	**Manufacturer/Model**	**Justification**	**Qty**	**Unit Cost Minus Tax (£)**	**Number Required (8) x Unit Cost (£)**
Monitor	AOC Q27G2S/BK 27" IPS QHD (2560×1440, 75 Hz, 1 ms MPRT, DP + HDMI)	AutoCAD can handle screens with a resolution as high as 3840×2160. Quad High Definition, which is 2560×1440, packs more pixels into the same space than 1080p. This makes it easier to read small text and details in drawings without needing to zoom in. Two 27-inch screens with QHD resolution: one used for the AutoCAD drawing area, and the other for LibreOffice and reference materials, following the standard setup for dual-monitor research work. An IPS panel ensures colours are displayed accurately and provides a wide viewing angle,	2	298	2,384

		making it ideal for technical drawings where shapes need to look correct from different angles. £149 per unit is a good value for a QHD IPS monitor; buying a high-end ultra-wide or 4K display would use up funds that could be better spent on main computer parts.			
Keyboard	Logitech MK120 wired USB keyboard	The spill-resistant design and quiet typing make it perfect for use in a lab setting.	1	16	128
Mouse	Logitech M90	Wired optical USB mouse; offers precise control suitable for CAD cursor work.	1	10	80
Total				£324	**£2,592**

Table 5. External Essentials for PC Design

4.2 Essential Software					
Software	**Manufacturer/Version**	**Justification**	**Qty**	**Unit Cost Minus Tax (£)**	**Number Required (8) x Unit Cost (£)**
Operating System	Microsoft Windows 11 Pro 64-bit	A Windows 11 Pro OEM license is a one-time fee paid per computer. OEM licenses cost much less than retail licenses, but they are legally connected to the hardware. AutoCAD 2026 needs a 64-bit version of Windows 11 or Windows 10, and Microsoft Windows 11 Pro 64-bit is one of the specified options. ARM processors are explicitly not supported.	8	135	1,080

		Windows 11 Pro is chosen because BitLocker Drive Encryption is a feature only available in the Pro version, and it's important for keeping CAD design data safe. Windows 11 Pro supports domain joining, linking PCs to the Active Directory on the Testing Centre domain controller at 172.16.1.54 in the network design. Remote Desktop is available in the Pro version for managing IT systems.			
CAD	Autodesk AutoCAD 2026 - full version	Annual subscription for each named user, which includes 3D capabilities and 7 toolsets.	8	1,490/yr	11,920
Office suite	LibreOffice 24.x	LibreOffice is free and open-source and fully supports Microsoft Office file formats such asbnxnb .docx, .xlsx, and .pptx, making it easy to share and exchange documents with other departments. Includes Writer for writing, Calc for spreadsheets, Impress for presentations, and Draw for vector graphics.	8	0	0
Total				£1,625	**£13,000**

Table 6. Essential Software for PC Design

4.3 PC Design Costing Summary

Category	**Qty**	**Cost (£)**
Hardware	1	1,334
Operating System	1	135
Subtotal	1	1,469
Incl. VAT @ 20%	1	1,762.80
Total PC design cost per machine		£1,762.80
Total cost savings per machine		**£237.20**

Table 7. Total Cost Estimate for PC Design

The PC design prioritises single-core CPU speed, which is key to how quickly AutoCAD responds. It includes 32 GB of DDR5 RAM, which Autodesk recommends. There's also a dedicated graphics card that supports DirectX 12 and has 8 GB of video memory, along with a quick NVMe SSD. These are the same specs Autodesk recommends for AutoCAD 2026.

Moreover, the PC lacks an optical drive because it's not needed. The software is provided through a subscription download, and all data is shared using the Testing Centre's network.

4.4 Comparison of Autocad Requirements vs Researcher's PC

PC Component	AutoDesk Recommendation	Flies & Welders PC	Status
CPU	3+ GHz base, 4GHz turbo	I7-4700: 5.4GHz boost, 20 cores	Exceeds requirement
RAM	32GB	32GB DDR5 dual channel 5600 MHz	Meets requirement
GPU VRAM	8GB	RTX 4060: 8GB GDDR6	Meets Requirement
GPU Bandwidth	106 GB/s	RTX 4060: 272 GB/s	Exceeds requirement
DirectX	DirectX 12	RTX 4060: DX12 Feature Level 12_0	Meets Requirement
Storage	16Gb - 20GB Solid State Drive (SSD)	1 TB NVMe PCIe 4.0	Exceeds requirement
Display Resolution	Upto 3840x2160	2560x1440 QHD IPS	Above minimum requirement
OS	Windows 11 64-bit	Windows 11 Pro 64-bit	Exceeds requirement

Table 8. AutoCAD system requirements vs Researchers PC Components

4.5 Security Features

To protect the Flies & Welders network, it's important to match the current hierarchical star topology and fibre backbone system with a full Defence-in-Depth approach (Cisco Networking Academy, 2019. By applying security across physical, technological, and administrative areas, the organisation can reduce risks such as spying by competitors and service disruptions.

4.5.1 Physical Controls

The Flies & Welders site has unique physical challenges, including a deep pool in the middle and a nearby electrical station. Physical security is the primary way to protect against hardware-related threats and prevent unauthorised access.

The current plan to use OM4 Multimode Fibre for the campus backbone provides protection against electromagnetic interference (EMI) from the on-site substation and heavy manufacturing equipment.

Because people with malicious intent can disrupt services by physically damaging equipment, all network cabinets in nearby locations, such as Security Buildings and Warehouses, should always remain locked. The industrial cabinets with an IP54 rating proposed for the Manufacturing Unit offer protection against dust and water, which are considered environmental risks, according to Cisco Networking Academy (2019).

Having a UPS (Uninterruptible Power Supply) in the Testing Centre and in important building cabinets helps protect against electrical issues such as sudden voltage changes or complete power failure. This ensures that security systems, such as gate access and IP cameras, continue to function even during a power outage.

4.5.2 Administrative Controls

Infrastructure devices, such as the Ubiquiti UniFi switches and the UDM-SE Core Router, need to be made more secure to prevent unauthorised changes to their settings.

Recommended Hardening Measures

Feature	Action for Flies & Welders	Purpose
Password Policy	Use longer passwords with 10 or more characters for every administrator account.	Prevents brute-force and dictionary attacks.
Secure Management	Disable Telnet and enable SSH (Secure Shell) on all managed switches.	Encrypts management traffic to prevent packet sniffing.
Service Control	Turn off any ports and services on the router that are not in use, such as HTTP or discovery protocols.	Reduces the attack surface available to intruders.
Banner Messages	Set up legal notice messages on every login screen.	Gives a legal reason to take action if there is a violation.

4.5.3 Technical Controls

As an aerospace company, it's really important to keep our special information and inventions safe. The following network-level features can be set up on the Fortinet FortiGate 80F Firewall and the core router.

Enhanced ACL Management: The current network setup already uses Access Control Lists (ACLs) to separate the Amenities building's browsing stations. This can be expanded to set up a DMZ (Demilitarised Zone) if the company later needs to host publicly accessible servers, helping ensure that external users can't access sensitive information about internal manufacturing or research and development.

AAA Framework: Using the Dell PowerEdge Domain Server to handle Authentication, Authorisation, and Accounting (AAA) will help maintain control of access and permissions in a single central place. This helps the IT team understand who is using the network, what actions they are allowed to take (such as keeping Warehouse staff from viewing Payroll information), and keeps a record of everything they do.

The UniFi GF IDF switch can require 802.1X port authentication (IEEE, 2010); computers must authenticate through Active Directory before they can access the network.

Malware Protection: The perimeter firewall can use Stateful Packet Inspection and Application Filtering to prevent known viruses, worms, and Trojan horses from reaching endpoint PCs.

Endpoint & Data Security: End devices, especially high-performance AutoCAD workstations in the Testing Centre, are often the primary targets for information theft. The four PCs in the Testing Centre on the ground floor test lab, located on the 172.16.1.0/26 subnet, are protected by the FortiGate firewall and can be separated using VLANs, so they cannot be accessed from the public internet.

Endpoint Patching: All computers running Windows 11 Pro should be configured to automatically receive security updates. According to the Cisco Networking Academy (2019), patching is the best method to reduce the risk of worm attacks.

Data Redundancy & Backups: The RAID-1 mirroring setup on the file server helps prevent data loss in the event of hardware failure. A formal security policy should also require regular off-site backups to guard against losing everything if the main location is damaged or a ransomware attack occurs.

Encryption: Using BitLocker Drive Encryption, which is part of Windows 11 Pro, on every researcher's computer will protect CAD design data if the hardware is stolen.

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